

Assignment on: Image editing using Photoshop;  
 Audio editing using Sound Forge;  
 Animation using Flash;  
 Video editing using Premier;  
 Authoring using Director;  
 Advanced Animation using 3D Max.

| 4 <sup>th</sup> Year 1 <sup>st</sup> Semester |                                  |              |          |         |         |   |    |        |       |
|-----------------------------------------------|----------------------------------|--------------|----------|---------|---------|---|----|--------|-------|
| Subject Code                                  | Subject Name                     |              | Category | Type    | Contact |   |    | Credit | Marks |
|                                               |                                  |              |          |         | L       | T | P  |        |       |
| IT/PC/B/T/411                                 | Machine Learning                 |              | PC       | Basic   | 3       | 0 | 0  | 3      | 100   |
| IT/PE/B/T/412A                                | Data Warehousing and Data Mining | Elective-III | PE       | Basic   | 3       | 0 | 0  | 3      | 100   |
| IT/PE/B/T/412B                                | IoT and Next Generation Networks |              |          |         |         |   |    |        |       |
| IT/PE/B/T/412C                                | Bio-Informatics                  |              |          |         |         |   |    |        |       |
| IT/PC/B/T/413                                 | Distributed Computing            |              | PC       | Basic   | 3       | 0 | 0  | 3      | 100   |
| IT/PC/H/T/414                                 | Network Security                 |              | PC       | Honours | 3       | 0 | 0  | 3      | 100   |
| IT/PC/B/S/411                                 | Machine Learning Lab             |              | PC       | Basic   | 0       | 1 | 3  | 2.5    | 100   |
| IT/PS/B/S/412                                 | Project                          |              | PS       | Basic   | 0       | 2 | 7  | 5.5    | 200   |
| IT/PS/B/S/413                                 | Seminar                          |              | PS       | Basic   | 0       | 1 | 2  | 2      | 100   |
|                                               | Internship                       |              | -        | -       | -       | - | -  | 0      | -     |
| Sub-Total                                     |                                  |              |          |         | 12      | 4 | 12 |        |       |
| Total                                         |                                  |              |          |         | 28      |   |    | 22     | 800   |

## (IT/PC/B/T/411) Machine Learning

**Introduction to Machine Learning:** A brief introduction to Machine Learning, Applications of ML, Design Perspective and Issues in ML, Supervised Learning, Unsupervised Learning, A Formal Learning Model and its representation, The Runtime of Learning

**Evaluating Hypotheses** – Estimating Hypotheses Accuracy, Basics of sampling Theory, Comparing Learning Algorithms; - Bayesian Learning – Bayes theorem, Concept learning, Bayes Optimal Classifier, Naïve Bayes classifier, Bayesian belief networks

**Artificial Neural Networks:** Early Models, Perceptron rule/multi-layer perceptrons, Artificial Neural Network, activation function, multi-layer neural network.

**Feedforward neural network:** Training Neural Network: Risk minimization, loss function Back-propagation I, Back-propagation II, Initialization, Training & Validation regularization, model selection, and optimization , Conditional Random Fields: Linear chain, partition function, Markov network, Belief propagation, Training CRFs, Hidden Markov Model, Entropy.

**Probabilistic Neural Network:** Hopfield Net, Boltzman machine, RBMs, Sigmoid net, Autoencoders.

**Modern Practical Deep Networks :** Perspectives and Issues in deep learning framework, Deep Feedforward Networks, Regularization for Deep Learning, Optimization for Training Deep Models, Convolutional Networks, Sequence Modeling: Recurrent and Recursive Nets

**Classification and Clustering :** Linear Models: Linear Classification, Logistic Regression , Linear Discriminant Analysis, Separating Hyper-plane Approaches, SVM –Formulation, SVM –Interpretation & Analysis, SVMs for Linearly Non Separable Data

**Reinforcement Learning:** Introduction to Reinforcement Learning, RL Framework , Solution Methods & Applications, Introduction to Immediate RL, Bandit Optimalities, Value function based methods

Policy Gradient Methods: Policy Search, REINFORCE, Contextual Bandits, Full RL Introduction Returns, Value Functions and MDPs

Dynamic Programming and Monte carlo methods: Lpi Convergence, Value Iteration, Policy Iteration, Dynamic Programming

Temporal Difference Methods : Off Policy MC, UCT, TD(0), Q-Learning

## **Elective-III**

### **(IT/PE/B/T/412A) Data Warehousing and Data Mining**

#### **Module-1**

**Data Warehousing and Business Analysis:** Basic concepts and Components of Data warehousing; Building a Data warehouse; Data Warehouse models-Data Mart; Metadata; Data Cube-multidimensional data model, Online Analytical Processing (OLAP), Multidimensional versus Multi-relational OLAP, OLAP Tools; Data Warehouse Design and Usage; Data Warehouse Implementation-OLAP Server Architectures-ROLAP versus MOLAP versus HOLAP; Data Cube Technology: Preliminary Concepts.

#### **Module-2**

**Introduction to Data Mining:** History of Data Mining, Definition, Knowledge Discovery vs. Data mining, Issues in data mining.

**Patterns of Data Mining:** Class Description, Characterization & Discrimination, Mining Frequent Patterns, Associations and Correlations, overview of cluster Analysis (preliminaries) and outlier analysis (preliminaries), Evolution Analysis.

**Data Preprocessing:** Data Summarization, data Cleaning, data Integration and Transformation, data Reduction, Data Discretization and Concept Hierarchy Generation.

#### **Module-3**

**Classical Data Mining approaches:** Scalable Frequent Itemset Mining Methods, The Apriori Algorithm, Mining various Kinds of Association Rules, Pattern-Growth Approach and Vertical Data Format for mining Frequent Itemsets, From Association mining to Correlation Analysis, Constraint based Association Mining, Mining in Multilevel and Multidimensional Space, Constraint-Based Frequent Pattern Mining, High-Dimensional Data mining with examples

#### **Module-4**

**Data Classification and Prediction:** Issues Regarding Classification and Prediction, Applications on- Decision Tree Induction, Bayesian Classification, and Rule-Based Classification; Classification by Association, Model Evaluation and Selection, Methods of Improving Classification Accuracy, Classification by Back-propagation , Classification Using Frequent Patterns, Multiclass Classification, Semi-Supervised Classification, Applications on K-Nearest Neighbor Classifier.

#### **Module-5**

**Cluster Analysis and outlier detection:** Types of data and requirements in Cluster Analysis, Partitioning methods, Application based Partitioning Methods in Large databases- From K-Means to CLARANS, Application of Hierarchical methods- BIRCH and Chameleon; Application of Density-Based Methods- DBSCAN, OPTICS, and DENCLUE; Grid based Methods- STING; and Clustering High Dimensional Data- CLIQUE; Types of Outliers, Challenges in Outlier Detection, Outliers Detection Methods, Statistical Approaches, Proximity-Based Approaches, Outlier Detection in High-Dimensional Data with examples.

## **Module-6**

**Overview of application oriented mining:** Mining Text Data, Mining Biological Data, Time Series Data etc.

### **(IT/PE/B/T/412B)    IoT and Next Generation Networks**

**Introduction:** Defining IoT, characteristics of IoT, challenges of IoT, Technologies leading to IoT, Functional blocks of IoT, M2M vs IoT. IoT Ecosystem: 7 layer model and protocols.

**IoT data link protocols:** Performance parameters, issues and challenges, IEEE 802.15.4, 802.11 AH, LoRaWAN, LTE-A, Z-wave, G.9959

**Network layer:** Routing protocols: RPL, CORPL, Encapsulation protocols: 6LoWPAN, IPv6 over G.9959.

**Transport and session layer:** Need for new protocols, MQTT, SMQTT, CoAP, XMPP

**IoT Management protocols:** IEEE 1905.1, Smart transducer interface.

**IoT Applications:** Home Automation, Smart manufacturing and smart factory, Surveillance applications, Smart parking, Intelligent traffic systems, Smart vehicles, Smart water management, IoT for smart cities.

**Next generation Networks:** MPLS, 5G Networks, MIMO, Vehicular Communication, SDN, Network slicing, D2D, Multi Radio Access Technologies

### **(IT/PE/B/T/412C)    Bio-Informatics**

**Fundamental of Cell Biology:** Definition. Basic concepts: protein and amino acid, DNA & RNA, Biological Sequence, structure and function, Genomes; Pattern recognition and prediction, Homology and Analogy.

**Bioinformatics databases:** Introduction, Type of databases, Nucleotide sequence databases: Primary nucleotide sequence databases, Secondary nucleotide sequence databases; Protein sequence databases, Sequence motif databases, Protein structure databases.

**Functional proteomics and genomics:** Mapping and Sequencing Genomes, Genetic Interactions, Protein profiling.

**Sequence alignment and database searching:** Introduction to database search, Algorithms issues in database search, Sequence database search, Single sequence alignments, Pair wise alignments: Scoring matrix, PAM, BLOSUM; Dynamic Programming, Heuristic methods: FASTA, BLAST; Multiple Sequence Alignments.

**Pattern Analysis:** Feature extraction, Classification: Linear classification, linear classification function and artificial neural, artificial neural networks, Support vector machines; Clustering: K-means clustering, hierarchical clustering; Dimensionality Reduction & Principal Component analysis: SDV, geometric interpretation of SDV, PLS method; Parametric Transformations: Hough transform, Generalized Hough transforms, geometric hashing;

**Evolutionary trees:** phylogeny: Ultrasonic trees, parsimony, Ultrametric problem, perfect phylogeny, phylogenetic alignment, connection between multiple alignment and tree construction

**Some advanced topics:** DNA Mapping and sequencing, Map alignment, Large scale sequencing and alignment, Shotgun, DNA sequencing, Sequence assembly, Gene predictions, Molecular predictions with DNA strings

**Markov chains and applications:** Machine Learning Methods, Hidden Markov models, Applications of HMM in gene identification

**Some Tasks in Computational Biology:** Fragment assembly, Sequence alignment, Gene finding, Promoter identification, Phylogenetic tree construction, Protein superfamily classification, Protein structure prediction, Protein folding

## **(IT/PC/B/T/413)**

## **Distributed Computing**

**Fundamental concepts:** Models, Issues, Complexity measures, proving correctness

**Clocks and Event Ordering:** Concept of clock in Distributed System, Limitation of Distributed System, Clock synchronization, Lamport's Logical Clock, Vector Clocks, Causal ordering of messages- Birman-Schiper Stephen Protocol, Schiper Egli Sandoz Protocol.

**Global state and snapshot recording algorithms:** System model, Snapshot algorithms for FIFO channels, Variations of Chandy-Lamport algorithm, Snapshot algorithms for non-FIFO channels, Snapshots in a causal delivery system, Monitoring global state, Necessary and sufficient conditions for consistent global snapshots, Finding consistent global snapshots in a distributed computation.

**Termination detection:** Introduction and issues

**Fundamental Algorithms:** Wave and Traversal Algorithms: Definition and use of wave algorithms, A collection of wave algorithms, Traversal algorithms, Leader Election algorithms: Introduction, Ring Networks, Arbitrary networks, The Korach-Kutten-Moran Algorithm.

**Distributed Mutual exclusion:** Permission based (Lamport, Ricart-Agarwala, Roucairol-Carvalho, Maekawa), Quorum-based mutual exclusion algorithms (Maekawa's algorithm, Agarwal-El Abbadi quorum-based algorithm), Token based (Suzuki Kasami, Raymond's Tree-based algorithm)

**Deadlock detection:** System model, Models of deadlocks, Knapp's classification of distributed deadlock detection algorithms, Mitchell and Merritt's algorithm for the single resource model, Chandy-Misra-Haas algorithm for the AND model, Chandy-Misra-Haas algorithm for the OR model, Kshemkalyani-Singhal algorithm for the P-out-of-Q model

**Distributed File System:** NFS, Google file systems

**Distributed Transactions:** The Management of Distributed Transactions, A Framework for Transaction Management, Supporting Atomicity of Distributed Transactions, Concurrency Control for Distributed Transactions, Architectural Aspects of Distributed Transactions;

**Distributed Concurrency Control:** Foundation of Distributed Concurrency Control, Distributed Deadlocks, Concurrency Control based on Timestamps, Optimistic Methods for Distributed Concurrency Control, Execution schedules;

## **(IT/PC/H/T/414)**

## **Network Security**

**Introduction to Network Security:** Overview - Define trust, weaknesses and vulnerabilities, Security Objectives; Understanding Vulnerabilities the Need for Security - Risk and Vulnerability, TCP/IP Suite Weaknesses, Buffer Overflows, Spoofing Techniques, Social Engineering; Understanding Defenses - Digital IDs, Intrusion Detection System, Encrypted Login, Firewalls, Reusable Passwords, Antivirus Software, Encrypted Files, Biometrics

**Security at the Application layer:** E-MAIL-E-mail Architecture; PGP – Scenarios, Key Rings, PGP Certificate, Key Revocation, PGP packets and Messages; S/MIME – MIME, S/MIME, Application of S/MIME

**Security at the Transport layer:** SSL Architecture, Four Protocols – Handshake, ChangeCipherSpec, Alert, Record, SSL Message formats, Transport Layer Security

**Security at the Network layer:** IPSec – Transport and Tunnel Mode, IPSec Security Association, Security protocols, Services Provided by IPSec, Security Policy, Internet Key Exchange (IKE) - Improved Diffie-Hellman Key Exchange, Phases and Modes, SA Algorithms, ISKAMP

**Firewalls:** Definition, Types of Firewalls - Circuit-Level Firewalls, Proxy Server Firewalls, Non-stateful Packet Filters, Stateful Packet Filters, Personal Firewalls (Packet Inspection Firewalls, Application Proxy Server: Filtering Based on Known Services, Virtual Private Network (VPN)

Firewalls, Small Office or Home (SOHO) Firewalls, NAT Firewalls), Configuration and Implementation of a Firewall, Improving Security Through the Firewall, Firewall Services and Limitations

**Access Control and Authorization:** Access Rights, Access Control Systems - Physical Access Control, Access Cards, Electronic Surveillance, Biometric, Event Monitoring; Authorization Systems – Centralize, Decentralized, Implicit, Explicit, Authorization Principles - Least Privileges, Separation of Duties; Web Access and Authorization

**Authentication:** Authentication Elements, Types of Authentication - Non-repudiable and repudiable; Authentication Methods - Password Authentication, Public Key Authentication, Remote Authentication, Anonymous Authentication, Digital Signatures-Based Authentication, Wireless Authentication

**IEEE 802.11 Wireless LAN Security:** Authentication – Pre-WEP authentication, Authentication in WEP, Authentication and Key agreement in 802.11i; Confidentiality and Integrity - Data Protection WEP, Data Protection TKIP and CCMP.

**Types of attacks, their analysis and mitigations:** phishing, wormhole, Sybil, DOS-DDOS, Intrusion Detection System

**Web Services Security:** Recapitulation : Technologies for Web Services, WS-Security - Token Types, XML Encryption, SML Signatures, SAML – Assertion Type, Creating/Communicating Assertions, Other Standard – WS-Trust, WS-Security Policy

## **(IT/PC/B/S/411) Machine Learning Lab**

Be familiar with necessary tools and languages for using machine learning techniques.

Assignments for finding the most specific hypothesis from a given set of training data samples.

Assignments on implementation of artificial neural networks (ANN) with back propagation and belief propagation, and using them to solve real world problems.

Assignments combining Hidden Markov Model and ANN.

Assignments using CNN and RNN to solve practical problems in areas such as game playing, sentence understanding.

Implementation of statistical classifiers for a sample training data set and to compute different metrics (such as accuracy, recall) of these classifiers.

Develop machine learning methods to cluster data and modify existing methods to solve the problem better.

Be familiar with tools for reinforcement learning (RL) (such as openAI gym or TensorLayer in TensorFlow) and assemble RL modules for tackling some interesting problems.

## **(IT/PS/B/S/412) Project**

Each student will have to carry out a project work for one year based on a suitable topic chosen in consultation with a teacher (the supervisor) concerned. This work will continue for two semesters in the final year. The marks for this subject will be awarded at the end of the second semester along with the marks obtained in (IT/PS/B/S/422).

## **(IT/PS/B/S/413) Seminar**

In this subject, each student will have to present a seminar on a topic in consultation with a teacher concerned and will be evaluated based on the content and format of his/her presentation. The marks for

this subject will be awarded at the end of the second semester along with the marks obtained in (IT/PS/B/S/423). The topic must be related to the broad domains of study in this course but the topic must not be chosen such that it is wholly contained in the syllabus of some subject in this course.

| 4 <sup>th</sup> Year 2 <sup>nd</sup> Semester |                                      |             |         |         |   |    |        |       |
|-----------------------------------------------|--------------------------------------|-------------|---------|---------|---|----|--------|-------|
| Subject Code                                  | Subject Name                         | Category    | Type    | Contact |   |    | Credit | Marks |
|                                               |                                      |             |         | L       | T | P  |        |       |
| IT/PC/H/T/421                                 | Data Science                         | PC          | Honours | 3       | 0 | 0  | 3      | 100   |
| IT/HS/B/T/422                                 | Management                           | HS          | Basic   | 3       | 0 | 0  | 3      | 100   |
| IT/PE/H/T/423A                                | Pattern Recognition and Applications | Elective-IV | PE      | 4       | 0 | 0  | 4      | 100   |
| IT/PE/H/T/423B                                | NLP and Text Mining                  |             |         |         |   |    |        |       |
| IT/PE/H/T/423C                                | Cyber Forensics and Security         |             |         |         |   |    |        |       |
| IT/PC/B/T/424                                 | Digital Image Processing             | PC          | Basic   | 3       | 0 | 0  | 3      | 100   |
| IT/PC/B/S/421                                 | Comprehensive Viva Voce              | PC          | Basic   | 0       | 0 | 3  | 1.5    | 100   |
| IT/PS/B/S/422                                 | Project                              | PS          | Basic   | 0       | 2 | 7  | 5.5    | 200   |
| IT/PS/B/S/423                                 | Seminar                              | PS          | Basic   | 0       | 1 | 2  | 2      | 100   |
| Sub-Total                                     |                                      |             |         | 13      | 3 | 12 |        |       |
| Total                                         |                                      |             |         | 28      |   |    | 22     | 800   |

## (IT/PC/H/T/421) Data Science

**Introduction to Data Science:** Basics of Data Science, The need for Business Analytics, Data Science Life Cycle, Different tools available for Data Science

**Statistical Inference:** Data exploration, Introduction to Research Methods, Central Tendency, Variability, Standardizing, Normal Distribution, Sampling Distributions, Hypothesis Testing, Estimation (Point estimate and Interval estimate), t-tests, ANOVA, Correlation, Regression, Chi-square Test

**Data Science Tools and Techniques:** Introduction of Hadoop file system and Map-reduce, Virtualization platforms, Cloud platforms;

Programming tools – R, Python, etc.

R: R packages and R Operators;

Python libraries: SciPy and sci-kitLearn, PyBrain, Pylearn2;

**Information Retrieval and Web Search:** Design issues related to IR, Working principle of Search Engine like Google, Well-known IR techniques used in various domains.

**Business Intelligence and Financial Analysis:** Understanding the BI process, Strategies of information gathering, the distinction between intelligence, information and data, Information asymmetry and competitive advantage, comparison with competitive intelligence, Comparison with business analytics, Success factors of implementation.

**Social Network and Sentiment Analysis:** Types of Networks- nodes, edges, adjacency matrix, one and two-mode networks, node degree, Random network models, Concepts: connected components, giant component, preferential attachment, Network centrality, betweenness, closeness, eigenvector centrality (+ PageRank), network centralization, Community: clustering, community structure. Introduction to Sentiment, Sentiment analysis with examples; Recommender System: Introduction to Recommender

Systems, Eliciting Ratings and other Feedback Contributions, Implicit Ratings, Taxonomy of Recommender Systems, Non-Personalized Recommenders Content-Based Recommenders, Methods of Filtering.

**Data Visualization:** Introduction to data visualization, Terminology. Basic Charts and Plots, Multivariate Data Visualization; Principles of Perception, Text Data Visualization, Interactivity and Animation; Temporal Data Visualization, Geospatial Data Visualization, Visualization Case Studies, Redesign Principles and Design Dimensionality

**Time Series:** Introduction of Time Series data, Time Series variables, Different components of Time Series data, Visualize the data to identify Time Series Components, Implement ARIMA model for forecasting, Exponential smoothing models, Identifying different time series scenario based on which different Exponential Smoothing model can be applied, Implement respective ETS model for forecasting

## **(IT/HS/B/T/422) Management**

**Principles of managements:** Concepts of management, development of scientific management, principles of Frederick Taylor & Henry Favol, functions such as planning, organizing staffing leading, motivating communicating controlling decision making span of control.

**Personal Management:** Meaning, functions of personal management, manpower planning, collective bargaining, wages & salary administration, labor welfare, training, trade unions, Industrial Factories Act, Industrial Boils Act, Trade Union Act.

**Plant Management:** Plant location, plant layout, types of maintenance such as break down, predictive & preventive maintenance, stores management, industrial safety, causes & cost of accidents, safety programs, production planning & control job, batch & process type of production.

**Marketing Management:** Definition & scope, selling & modern concepts of marketing, market research, new product development, product life cycle, product lunching, sales promotion, pricing, channels of distribution, advertising market segmentation, marketing mix.

**Material Management:** Importance of Materials Management, Classification, Codification, Forecasting, Necessity of Inventory.

**Financial Management:** Sources of finance, financing organization, types of capital, elements of costs & allocations of indirect expenses, cost control, break even analysis, budges & budgetary control, equipment replacement policy, make or buy analysis, balance sheet, ratio analysis, profit & loss statement.

## **Elective-IV**

### **(IT/PE/H/T/423A) Pattern Recognition and Applications**

**Module I: Introduction and mathematical preliminaries:**

Introduction to Pattern recognition, its Applications and Examples, Basics of Linear Algebra and Vector Spaces, Basics of probability theory, Estimation theory, Matrices, Metric and non-metric proximity measures, Distance Matrices, Decision boundary, Decision Region, Difference between Supervised and unsupervised learning, Training set, test set; standardization and normalization.

**Module II: Classification techniques:**

Supervised Learning, Bayes decision rule, Errors in probability theory, Linear Discriminant Function, Non-linear Decision Boundaries, Minimum Distance Classifier, K-NN Classifier, Fisher's LDA, Support Vector Machine, Bagging, Boosting / AdaBoost, Regression & Classification, Logistic Regression, Decision Trees.

**Module III: Clustering techniques:**

Unsupervised Learning, Basics of Clustering; similarity / dissimilarity measures; clustering criteria, Minimum within cluster distance criterion, K-means algorithm, Single linkage and complete linkage algorithms, MST, K-medoids, DBSCAN

**Module IV: Feature Extraction & Selection:**

PCA, Kernel PCA, Branch and bound algorithm, sequential forward / backward selection algorithms, Probabilistic separability based criterion functions, interclass distance based criterion functions.

**Module V: Advancement and application of pattern recognition:**

Applications of machine learning

and pattern recognition in solving problems drawn from domains of audio classification, biometric recognition, image retrieval, bio-informatics etc.

## **(IT/PE/H/T/423B) NLP and Text Mining**

**NLP****Basics of NLP:**

Human languages, models, ambiguity, processing paradigms; Phases in natural language processing applications; Text representation in computers, encoding schemes, Regular Expressions, Finite State Automata, Word Recognition, lexicon, Word Tokenization, Normalization, Sentence Segmentation, Named Entity Recognition, Multi Word Extraction, Spell Checking – Bayesian Approach, Minimum Edit Distance, Context Free Grammar and Parsing

**Morphology and Finite State Transducers**

Inflectional and Derivational Morphology, Finite State Morphological Parsing, The Lexicon and Morphotactics, Morphological Parsing with Finite State Transducers, Combining FST Lexicon and Rules, Orthographic Rules and Finite State Transducers, Porter Stemmer

**Introduction to N-grams**

Simple (Unsmoothed) N-grams, Smoothing – Add-One Smoothing, Witten-Bell Discounting, Good-Turing Discounting; Backoff, Deleted Interpolation, N-grams for Spelling, Entropy, Cross Entropy for comparing models.

**Hidden Markov Models and Parts Of Speech Tagging**

Markov Chain, Hidden Markov Models, Forward Algorithm, Viterbi Algorithm, Part of Speech Tagging: Rule based POS tagging, Stochastic POS tagging, Transformation based tagging, Multiple tags and multiple words, Handling of unknown words, named entities, multi word expressions.

**Semantics analysis**

Meaning representation, Syntax- Driven Semantic analysis, Relation among lexemes and their senses- Homonymy, Hyponymy, Synonymy. Lexical semantics, WordNet: A database of lexical relations, Word Sense Disambiguation, Computational Lexical Semantics: Thesaurus based and Distributional Word Similarity.

## **Text Mining**

### **Information Retrieval and Extraction**

Boolean Retrieval, Term weighting and vector space model, Term-document incidence, The Inverted Index, Query Optimization, Phrase Queries, Ranked Retrieval – Term Frequency – Inverse Document Frequency based ranking, Zone Indexing, Query term proximity, Cosine ranking, Combining different features for ranking, Search Engine Evaluation, Relevance Feedback

### **Question Answering and Textual Entailment**

Open and Restricted Domain Question-Answering; Question Classes and Processing; Context and Data Source; Extraction and Formulation of Answers; Web exploitation: Real Time, Multilingual and Interactive QAs; Textual Entailment: Learning to recognize features; Textual Inference and Semantic Entailment.

### **Text Summarization**

Automatic Summarization: Factors and Directions, Extraction, Abstraction, Maximum Entropy and Aided Summarization; Keyphrase Extraction; Supervised and Unsupervised Learning for Sentence Compression, Sentence Fusion, Document Summarization: Single and Multidocument.

### **Performance Measure**

Precision, Recall and F-Measure, Confusion Matrix; MAP, MRR, N-gram Co-Occurrence Statistics for Summarization; Measures: Intrinsic (Coherence, Informativeness, Utility Method, Content Similarity, BLEU and ROUGE)

## **(IT/PE/H/T/423C) Cyber Forensics and Security**

Introduction to Computer Forensics: computer crimes, evidence, extraction, preservation, etc.

Overview of hardware and operating systems: structure of storage media/devices; windows/Macintosh/Linux -- registry, boot process, file systems, file metadata.

Data recovery: identifying hidden data, Encryption/Decryption, Steganography, recovering deleted files.

Digital evidence controls: uncovering attacks that evade detection by Event Viewer, Task Manager, and other Windows GUI tools, data acquisition, disk imaging, recovering swap files, temporary & cache files

Computer Forensic tools: Encase, Helix, FTK, Autopsy, Sleuth kit Forensic Browser, FIRE, Found stone Forensic ToolKit, WinHex, Linux dd and other open source tools.

Network Forensic: Collecting and analyzing network-based evidence, reconstructing web browsing, email activity, and windows registry changes, intrusion detection, tracking offenders, etc.

Mobile Network Forensic: Introduction, Mobile Network Technology, Investigations, Collecting Evidence, Where to seek Digital Data for further Investigations, Interpretation of Digital Evidence on Mobile Network.

Software Reverse Engineering: defense against software targets for viruses, worms and other malware, improving third-party software library, identifying hostile codes-buffer overflow, control hijacking attacks, provision of unexpected inputs, Exploitation techniques and fuzzing- Finding vulnerabilities and exploits.

Dealing with bad (legacy) application code: Sandboxing and Isolation. Least privilege, access control, operating system security

Malicious Software and Software Security- Malicious Web, Internet Security Issues, Types of Internet Security Issues, Computer viruses, Spyware, Key-Loggers, Secure Coding, Electronic and Information Warfare.

Mobile platform security models- Android, iOS Mobile platform security models, Detecting Android malware in Android markets.

Basic web security model: Browser content, Document object model (DOM), Same-origin policy.

Web Application Security: SQL injection, Cross-site request forgery, Cross-site scripting, Attacks and Defenses, Generating and storing session tokens, Authenticating users, The SSL protocol, The lock icon, User interface attacks, Pretty Good Privacy.

Computer crime and Legal issues: Intellectual property, privacy issues, Criminal Justice system for forensic, audit/investigative situations and digital crime scene, investigative procedure/standards for extraction, preservation, and deposition of legal evidence in a court of law.

## **(IT/PC/B/T/424)      Digital Image Processing**

**Fundamentals:** Overview of Image Processing System, image digitization (sampling and quantization), basic relationship between pixels, Fundamentals of Color image and Color Models

**Image Transform:** Fourier transform (1D, 2D) and its properties, FFT, DCT, Hadamard transform, Karhunen – Loeve transforms.

**Image Enhancement:** Spatial Domain methods- Contrast Intensification (linear and nonlinear stretching), Histogram equalization, Histogram specification, Spatial filtering: Smoothing (linear, order-statistic), sharpening filters (Laplacian), Frequency domain filters : Smoothing, Sharpening filters, Homomorphic filtering.

**Image Segmentation:** Detection of discontinuities (point, line & edge), Edge Linking and Edge Following by Local Processing, Hough Transform, Region Extraction by Pixel based Approach (Thresholding, Choice of Feature, Optimum Threshold etc.), Region Extraction by Region based Approach (Region Growing, Splitting, Merging, Split and Merge).

**Morphological Image Processing:** Preliminaries, Erosion and Dilation, Opening and Closing, Some Basic Morphological Algorithms: boundary extraction, region filling.

**Image representation & description:** Representation: Chain Codes, Polygonal Approximation, Skeletons, medial axis transform, thinning, Descriptor: shape number, Fourier descriptor, statistical moments, Geometrical Attributes (perimeter, area, etc.), Texture Descriptor (GLCM).

**Image Compression:** Objective, Feasibility, performance measure, lossy/lossless compression, Huffman coding, Arithmetic coding, block truncation coding, vector quantization, JPEG.

## **(IT/PC/B/S/421)      Comprehensive Viva Voce**

Each student will have to attend a viva-voce examination before a panel of experts drawn from people inside and outside Jadavpur University. Questions will be asked from any topic taught as a part of this curriculum.

## **(IT/PS/B/S/422) Project**

Each student will have to carry out a project work for one year based on a suitable topic chosen in consultation with a teacher (the supervisor) concerned. This subject is in continuation to the project work

started in the first semester of the final year in the subject (IT/PS/B/S/412). The marks for this subject will be awarded at the end of the second semester along with the marks obtained in (IT/PS/B/S/412).

### **(IT/PS/B/S/423) Seminar**

In this subject, each student will have to present a seminar on a topic in consultation with a teacher concerned and will be evaluated based on the content and format of his/her presentation. The topic must be related to the broad domains of study in this course but the topic must not be chosen such that it is wholly contained in the syllabus of some subject in this course.

## **Open Electives (Catered by Dept. of IT)**

### **To be taught in odd Semester**

#### **(FET/OE/B/T/0XX) Database Management Systems Basics**

**Introduction:** History of Evolution of DBMS and advantages over traditional file system, Three-schema architecture of DBMS and Data Independence. Introduction to DDL and DML. Ideas about different kind of users of DBMS and available databases in market.

**Data Model:** Introduction to Relational data model; Keys, Entity-Relationship Model, Relational Algebra

**Database Design:** Different types of dependencies, Normalization

**SQL:** Introduction to SQL, Application development using SQL

**Data Storage and Querying:** Physical data structure, Evaluation of Relational Algebra expressions

**Transaction Management:** Transaction Processing, Concurrency control and Recovery management, Transaction Model Properties and State, Serializability, Lock-based, Two-phase Locking.

### **To be taught in even Semester**

#### **(FET/OE/B/T/0XX) Object Oriented Programming Fundamentals**

**Basic Concepts of Object Oriented Programming (OOP)** – Objects, Classes and Message Passing, Differences between conventional and Object-Oriented programming, advantages and disadvantages of OOP.

**Enhancements over Procedural Languages:** New data types. Function Overloading and its resolution, New dynamic allocation and de-allocation methods.

**Classes and Objects:** Notions of abstraction, encapsulation, information hiding and modularity. Instantiation and initialization of objects; constructor and destructor, Access Specification, Functions and methods; self reference of objects, Copy Constructor, Static Members.

**Inheritance and Polymorphism:** Basic concept of IS-A relationship. “protected” access specification, Initialization and de-initialization of derived objects, Polymorphic method calls and dynamic binding. Abstract methods and classes. Examples, IS-A versus IMPLEMENTED-IN-TERMS-OF relationship, Multiple Inheritance, The problem of multiple occurrence of the same base and its solution.

**Operator Overloading (C++):** Fundamental ideas, Examples of overloading with arithmetic, relational operators. Overloading of unary operators others.

**Basic I/O and File I/O.**

**Exception handling:** The idea of exception handling and its superiority over traditional error handling. Semantics of *try-catch* blocks and *throw*.